MRITYUNJAY JHA

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Education

Delhi Technological University Aug 2023 - May 2027
B.Tech in Computer Science and Engineering *CGPA: 8.00*

Govt Boys Sr Sec School, Dwarka Sec-3 2020 - 2022
Senior Secondary (XIIth Grade) *Grade: 86%*

Ring Midways Sr Sec Public School 2019 - 2020
Secondary (Xth Grade) *Grade: 94.8%*

Experience

QuickIntell, Remote Jun 2026 - Jul 2026
Software Engineer Intern *Certificate*

- Architected a **claims processing MVP platform**, automating the end-to-end insurance lifecycle and reducing manual review overhead by **80%** via **Straight-Through Processing (STP)**.
- Integrated a secure ingestion pipeline using **Gmail Connectors** and **OAuth2**.
- Developed an AI-driven medical coding system using **RAG (Retrieval-Augmented Generation)** and **vector databases** to map complex clinical narratives to ICD-10 codes, achieving **over 95% accuracy** in diagnosis extraction.

Projects

ALASKA | *React, Node.js, Express, PostgreSQL, Prisma, Redis, PgBouncer, Socket.IO, Docker* GitHub

- Engineered a **route-based social networking platform** that enables **real-time discovery**, connecting users who share overlapping metro commutes.
- Developed an interactive custom metro route builder using **Mapbox GL JS** and **Dijkstra's algorithm**, computing optimal shortest paths across a network of 250+ interconnected nodes in real-time.
- Optimized subpath matching by implementing a **Suffix Graph**, reducing time complexity from quadratic to $\mathcal{O}(N \times L)$ and offloading generation to the client to eliminate server-side preprocessing overhead.
- Built a scalable **real-time chat system** utilizing a **Gateway-Worker-Queue pattern** with **Socket.IO and Redis**, decoupling connections from database writes to ensure message delivery during high traffic spikes.

CORTEX | *Python, PyQt6, Pydantic, NetworkX, LLM, PyAutoGUI* GitHub

- Built an **OS-level automation engine** using **PyAutoGUI** to interact with complex EHR systems, enabling seamless data entry workflows across iframe-heavy web portals and legacy desktop apps where standard DOM scraping fails.
- Transformed script-based bot creation into a visual workflow by developing a custom **drag-and-drop node interface** using **PyQt6 and NetworkX**, allowing non-technical users to visually build complex **automation pipelines (DAGs)**.
- Reduced UI-update breakages by integrating **self-healing image locators** powered by **Vision LLMs**, enabling the system to dynamically identify and interact with elements despite layout shifts or varying screen resolutions.

Technical Skills

Languages: C/C++, Python, JavaScript, TypeScript, Bash, HTML/CSS

Web & Backend APIs: React.js, Node.js, Express.js, REST APIs, WebSockets

Databases & ORM: PostgreSQL, MongoDB, MySQL, Prisma, ChromaDB

DevOps: Git, GitHub, GitHub Actions, Docker, AWS, Cloudflare Workers, Linux

Coursework: Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks

Achievements

Knight on LeetCode with 700+ problems solved, earning 15 badges and achieving a peak rating of 1860.

Specialist on Codeforces, participating in 15+ contests and reaching a maximum rating of 1410.